

Pakistan Artificial Intelligence Builders (PAIB)

Professional Course & Internship (3 Months)

Program: Generative AI, Agentic AI & AI Automation

Duration: 3 Months (12 Weeks)

Mode: Physical / Online

Internship Objective

This professional program is designed to transform beginners into AI Developers capable of building modern AI-powered applications, intelligent chatbots, AI agents, Retrieval-Augmented Generation (RAG) systems, AI automation workflows, and autonomous AI employees using today's leading AI technologies.

Participants will gain practical, hands-on experience with OpenAI, Claude, Gemini, OpenClaw, NanoClaw, Next.js, n8n, Neon PostgreSQL, GitHub, Vercel, and modern AI development best practices.

Module 1 – Artificial Intelligence Fundamentals (Week 1)

- History and Evolution of AI
- ANI, AGI & ASI
- Human Employees vs AI Employees
- AI Product Development Lifecycle
- AI Career Roadmap
- Future of AI
- AI Ethics
- AI Business Opportunities
- Practical: AI tools, AI accounts, GitHub & VS Code setup

Module 2 – Prompt & Context Engineering (Week 2)

- Prompt Engineering
- Context Engineering
- Loop Engineering
- AI Reasoning
- Chain of Thought
- Few-shot Prompting
- System Prompts
- Prompt Optimization
- Vibe Coding
- AI-Assisted Development
- Projects: Prompt Library, AI Assistant, AI Resume Generator

Module 3 – Modern AI-Powered Full-Stack Development (Week 3)

- Node.js
- npm
- VS Code
- Git & GitHub
- TypeScript
- Next.js
- React Fundamentals
- Tailwind CSS
- Deployment with Vercel

- Project: AI Portfolio Website

Module 4 – Chatbot Development (Week 4)

- Traditional Chatbots
- AI Chatbots
- RAG (Embeddings, Vector Databases, Retrieval, Chunking, Knowledge Bases, PDF Chatbots)
- Projects: Customer Support Chatbot, AI Customer Assistant, Company Knowledge Assistant

Module 5 – AI Agent Development (Week 5)

- AI Agent Components
- Planning
- Memory
- Reasoning
- Tool Calling
- Multi-Agent Systems
- AI Employees
- Autonomous Workflows
- Projects: Research, Sales, HR & Personal Assistant Agents

Module 6 – OpenClaw & NanoClaw Development (Week 6)

- Environment Setup
- Installation
- Configuration
- Customization
- Deployment
- Projects: HR, Marketing, Customer Support & Research AI Employees

Module 7 – AI Automation with n8n (Week 7)

- Automation Fundamentals
- n8n
- AI Workflows
- Webhooks
- APIs
- Email Automation
- WhatsApp Automation
- CRM Automation
- Lead Generation
- Projects: Gmail Bot, WhatsApp Bot, AI Email Assistant, CRM Automation

Module 8 – AI Product Development (Week 8)

- AI Websites
- Web Applications
- AI Chatbots
- RAG Applications
- AI Agents
- AI Automation
- AI SaaS
- AI Dashboards
- Final Capstone Project

Internship (Month 3)

- Work on real client projects including AI Chatbots, RAG systems, AI Agents, OpenClaw, NanoClaw, AI Employees, AI Automation, LMS, CRM, HR Systems, AI SaaS products, and AI Dashboards.
- Professional GitHub workflow and collaboration.

Learning Outcomes

- Build Traditional & AI Chatbots
- Build RAG Applications
- Build AI Agents
- Build AI Employees
- Deploy OpenClaw & NanoClaw
- Develop AI Automation with n8n
- Build AI-powered Full-Stack Applications
- Deploy Production-Ready Applications
- Work in Professional Development Teams
- Build Commercial AI Products
- Start Freelancing
- Launch an AI Startup

Technologies Covered

ChatGPT, OpenAI API, Claude API, Gemini API, OpenRouter, OpenClaw, NanoClaw, Next.js, React, TypeScript, Tailwind CSS, Node.js, Git, GitHub, Neon PostgreSQL, Prisma, n8n, Vercel, MCP, RAG, and Vector Databases.

Recommended Paid Services

- ChatGPT Plus – Custom GPTs, GPT-5 and advanced features – **\$20/month**
- OpenAI API – Estimated budget: **~\$50/month**
- Google Gemini API – Estimated budget: **~\$50/month**
- Claude Pro – **\$20/month**
- GitHub Pro – **\$4/month**
- Vercel Pro – **\$20/month**
- Neon PostgreSQL – From **~\$19/month**
- n8n Cloud – From **\$24/month**
- Cloudinary – Paid plans from **~\$25/month**
- WhatsApp Business API – Estimated **~\$50/month**